Functional Annotation of MlaD (Q88P92 / PP_0960) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

MlaD (UniProt **Q88P92**, ordered locus **PP_0960**) of *Pseudomonas putida* KT2440 is the **inner-membrane, phospholipid-binding subunit of the Mla ("Maintenance of Lipid Asymmetry") ABC transporter system**. It is **not an enzyme**: rather, it is a **substrate-binding / substrate-shuttling adapter** that binds **glycerophospholipids** and forms a **homohexameric ring (an MCE — Mammalian Cell Entry — domain ring)** anchored to the cytoplasmic (inner) membrane by a single N-terminal transmembrane helix, with the functional ring projecting into the **periplasm**. As the "D" component of the inner-membrane **MlaFEDB** ABC transporter complex, MlaD accepts and delivers phospholipids to/from the periplasmic shuttle protein **MlaC** and channels them, in an **ATP-dependent** manner, toward the MlaE permease. The physiological role of the complete OmpC/F–Mla pathway is the **retrograde (outer-membrane → inner-membrane) transport of mislocalized phospholipids**, which preserves the lipid asymmetry and permeability-barrier function of the Gram-negative outer membrane.

In *P. putida* specifically, *mlaD* (PP_0960) is co-transcribed with *mlaF* (ATPase, PP_0958) and *mlaE* (permease, PP_0959) in a contiguous operon and sits immediately upstream of the ***ttg2*** toluene-tolerance genes; this ties MlaD to *P. putida*'s hallmark **organic-solvent (toluene) tolerance** and outer-membrane barrier maintenance [PMID 9658016].

> **Identity confirmation.** The target's gene symbol (*mlaD*), locus (PP_0960), and domain architecture are fully self-consistent: InterPro/Pfam annotations **PF02470 (MlaD)**, **IPR030970 (ABC_MlaD)**, **IPR003399 (Mce/MlaD)**, and **IPR052336 (MlaD phospholipid transporter)** unambiguously place Q88P92 in the MlaD/MCE family. The Mla system is conserved in **all Gram-negative bacteria**, so functional inference from the intensively studied *E. coli* orthologue is well justified. No gene-symbol ambiguity was encountered.

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2. Background: The Mla Pathway

The outer membrane (OM) of Gram-negative bacteria is a uniquely **asymmetric bilayer** — lipopolysaccharide (LPS) in the outer leaflet and glycerophospholipids (GPLs) in the inner leaflet — and this asymmetry is what makes the OM an effective barrier against detergents, bile salts, and antibiotics. When phospholipids aberrantly accumulate in the *outer* leaflet of the OM, the barrier is compromised. The **Mla system** counteracts this by removing those mislocalized phospholipids and returning them to the inner membrane (IM) [PMID 34753108; 34873038].

The pathway comprises **three physically separated assemblies** bridged across the cell envelope [PMID 39080293]:

Location	Component(s)	Role
Outer membrane	**MlaA–OmpC/OmpF**	Extracts GPLs from the OM outer leaflet
Periplasm	**MlaC**	Soluble lipid-binding shuttle protein
Inner membrane	**MlaFEDB** ABC transporter (MlaF = ATPase; MlaE = permease; **MlaD** = substrate-binding ring; MlaB = STAS regulatory subunit)	Receives lipids at the IM; ATP-driven

MlaD is the **D subunit of the inner-membrane MlaFEDB complex**, which has been proposed to be the founding member of a **structurally distinct ABC transporter superfamily** [PMID 39080293; 35981415].

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3. Primary Function of MlaD

3.1 Molecular function — phospholipid-binding adapter (not a catalyst)

MlaD is described as a **membrane-associated solute-binding protein (SBP)** that "help[s] the transport of phospholipids (PLs) between the outer and inner membranes of Gram-negative bacteria" [PMID 38347327]. It is the **"unique IM-associated periplasmic solute-binding protein"** of the Mla system [PMID 34196044]. Unlike a classical enzyme, MlaD does not catalyze a chemical reaction; its function is to **bind glycerophospholipids and hand them off** between transport components. Its substrate is therefore **glycerophospholipid** (common bacterial species such as phosphatidylethanolamine and phosphatidylglycerol); the related shuttle MlaC is polyspecific and can bind two phospholipids in its pocket [PMID 36084896].

3.2 Substrate specificity

The transported cargo is **glycerophospholipid**, i.e., the diacyl-glycerophospholipids of the bacterial membrane. The system is thought to be **polyspecific** for the abundant membrane GPLs rather than selective for a single headgroup [PMID 36084896]. MlaD itself provides a hydrophobic conduit for the acyl chains (see §4).

3.3 Role in the transport reaction / directionality

- **MlaD couples periplasmic lipid delivery to ATP-driven translocation.** The MlaFEDB complex "functions via an unknown mechanism," and structures in apo, phospholipid-bound, ADP-bound and AMP-PNP-bound states identify residues that "recognize and transport phospholipids" [PMID 33199922].
- **Direction is retrograde (OM → IM).** In vitro point-to-point transfer assays show the OmpC-Mla system removes aberrantly localized PLs from the OM and transports them to the IM, and that **ATP binding/hydrolysis "disrupts lipid-binding equilibrium to drive retrograde transport"** [PMID 34873038]. (Some studies discuss possible bidirectionality, but the consensus physiological role is retrograde.)
- **MlaD is the acceptor node for the MlaC shuttle.** "Mla uses a shuttle-like mechanism to move lipids between the MlaFEDB inner membrane complex and the MlaA-OmpF/C OM complex, via a periplasmic lipid-binding protein, MlaC. **MlaC binds to MlaD**" [PMID 37100290]. Structures of the MlaC–MlaD complex show **≥2 MlaC molecules can dock on one MlaD hexamer**, and identify the **MlaD β6–β7 loop as essential** for MlaC–MlaD function; phospholipids pass **between the C-terminal helices of the MlaD hexamer to reach the central pore** [PMID 39080293].

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4. Structure and Structural Role

MlaD is the structural centerpiece linking the periplasmic and inner-membrane steps:

- **Protomer fold.** Each MlaD protomer has an **N-terminal seven-stranded β-barrel (the MCE/"MlaD domain")** plus a **C-terminal α-helical domain (HD)** [PMID 38347327]. An N-terminal single transmembrane helix (not part of the crystallized periplasmic region) anchors the protein in the inner membrane.
- **Hexameric ring.** Six protomers oligomerize into a **homohexameric ring with a continuous, largely hydrophobic central channel of variable diameter** [PMID 38347327]. Notably, the **C-terminal helical domain — not the β-barrel MCE domain itself — drives oligomerization** [PMID 38347327].
- **Functional pore loop.** A **conserved C-terminal "pore loop"** lining the central channel is functionally crucial for trafficking hydrophobic molecules; the MlaD domain is enriched in glycine and hydrophobic residues and lacks cysteines [PMID 34196044].
- **Lipid conduit.** The hydrophobic central pore, fed by lipids passing between the C-terminal helices, forms the pathway by which acyl chains are shielded from the aqueous periplasm en route to/from the MlaE permease [PMID 39080293].

This hexameric-MCE-ring architecture is the defining hallmark of the **MCE protein family**, which also includes the multi-ring PqiB and LetB/YebT transporters; YebT (an MlaD homologue) forms an elongated stack of MCE rings spanning between IM and OM [PMID 31870848]. In the diderm Firmicute *Veillonella parvula*, an **elongated MlaD even forms a complete transenvelope bridge** with its own OM β-barrel, and phylogenomics indicates **MlaEFD constitute the ancestral functional core** of the Mla system [PMID 37993432].

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5. Subcellular Localization

- **Inner (cytoplasmic) membrane, periplasmic-facing.** MlaD is an **IM-associated protein**; it is tethered to the inner membrane by an N-terminal transmembrane segment, with the **hexameric MCE ring projecting into the periplasm** atop the MlaF (ATPase) / MlaE (permease) / MlaB core [PMID 34196044; 35981415].
- **Site of function.** MlaD carries out its lipid hand-off at the **periplasmic face of the inner membrane**, receiving/delivering lipids from the soluble periplasmic shuttle MlaC and passing them to the transmembrane MlaE permease [PMID 37100290; 39080293].

6. Pathway Context and Physiological Consequences

- **Pathway.** MlaD operates within the **OmpC/F–Mla retrograde phospholipid-trafficking pathway**, functionally connecting the OM (MlaA-OmpC/F) → periplasm (MlaC) → IM (MlaFEDB). The energy source is **ATP hydrolysis by MlaF** at the IM [PMID 34873038; 35981415].
- **Opposing/partner systems.** Genetic evidence across lineages shows the Mla system has **opposite trafficking function to the TamB/AsmA-family (anterograde) systems**, consistent with Mla being retrograde [PMID 37993432].
- **Consequences of loss.** Disruption of the Mla pathway produces phenotypes of **compromised OM lipid asymmetry**: increased outer-membrane permeability, sensitivity to detergents/antibiotics, and **outer-membrane vesicle blebbing**; in pathogens the pathway contributes to **virulence** [PMID 39080293; 37993432]. These effects trace specifically to the failure to clear surface-exposed phospholipids, rather than to broad pleiotropy.

7. *P. putida* MlaD (Q88P92): Organism-Specific Sequence Evidence

Direct analysis of the actual Q88P92 sequence (retrieved from UniProt; locus PP_0960; RefSeq WP_010952152.1) provides organism-specific confirmation of the annotation, beyond pure orthology:

- **161 aa, 16.95 kDa — a canonical single-MCE-domain MlaD.** Q88P92 has one Mce/MlaD domain (Pfam **PF02470**, residues **39–116**) — the short, single-ring MlaD type (like *E. coli* MlaD ~183 aa), **not** the elongated multi-ring/transenvelope MlaD (YebT/*V. parvula* type). This is consistent with assembly into the standard inner-membrane MlaFEDB complex.
- **Single N-terminal transmembrane anchor.** Kyte–Doolittle hydropathy (window 19) shows one strong hydrophobic segment (peak score **2.86**, residues **~9–31**; `GVGLFLLAGILALLLLALRVSGL`), the predicted inner-membrane anchor helix, with the MCE domain following in the periplasm — exactly the topology required for MlaFEDB.

- **Diagnostic composition.** The protein is **cysteine-free (0 Cys)** and **glycine-rich (9.9%)**, matching the reported MlaD-domain signature ("abundance of glycine and hydrophobic residues and the lack of cysteine residues") [PMID 34196044].
- **Database annotations converge on the same function.** UniProt cross-references assign **GO:0005543 (phospholipid binding)** and **GO:0005548 (phospholipid transporter activity)**, TIGRFAM **TIGR04430 "OM_asym_MlaD"**, eggNOG **COG1463**, and PANTHER **"Intermembrane phospholipid transport system binding protein MlaD" (PTHR33371:SF4)**. {{figure:mlaD_hydropathy.png|caption=Kyte-Doolittle hydropathy profile and domain map of P. putida MlaD (Q88P92). A single strong N-terminal hydrophobic segment (peak ~2.86, residues ~9-31) corresponds to the lone inner-membrane transmembrane anchor helix; a single Mce/MlaD domain (PF02470, residues ~39-116) follows in the periplasm, with a C-terminal region (~117-161). This canonical single-anchor, single-MCE-domain topology matches the E. coli MlaD architecture and rules out the elongated transenvelope MlaD type.}}

Evidence base for the functional assignment

- **Directly experimental (E. coli / general Mla):** cryo-EM and crystal structures of MlaFEDB and MlaD (apo and lipid/nucleotide-bound), reconstituted proteoliposome transport assays, in vivo complementation and permeability assays, deep mutational scanning of MlaC, and MD simulations [PMID 33199922; 34873038; 39080293; 37100290; 38347327].
- **Bioinformatic / evolutionary:** MlaD-domain profiling, pore-loop conservation, phylogenetics, and the identification of MlaEFD as the ancestral core [PMID 34196044; 37993432]; plus the Q88P92-specific sequence analysis above.
- **Pathway conservation in non-model Gram-negatives:** the Mla pathway is essential for the intrinsic antimicrobial resistance and OM barrier of *Burkholderia cepacia* complex species, demonstrating the pathway's physiological importance beyond *E. coli* [PMID 29986943].
- **Caveat:** No *P. putida*-specific biochemical/genetic study of PP_0960 itself was found. The functional assignment rests on (i) the exact diagnostic domain set, (ii) organism-specific sequence features (single TM + single MCE domain, Cys-free/Gly-rich), and (iii) strong orthology to experimentally characterized MlaD. Direct measurement in *P. putida* remains a gap (see §9).

7a. Genomic Context in *P. putida*: the *mlaFED* Operon and Link to Solvent Tolerance

Inspection of the loci flanking PP_0960 in the KT2440 genome shows that MlaD is encoded within a **contiguous *mla* operon** and embedded in a region devoted to outer-membrane/LPS homeostasis:

Locus	Gene	Product (UniProt)
PP_0955	*lptC*	LPS export system protein LptC
PP_0956	*kdsC*	Kdo-8-phosphate phosphatase (LPS biosynthesis)
PP_0957	*kdsD*	Arabinose-5-phosphate isomerase (LPS biosynthesis)
PP_0958	***mlaF***	Intermembrane phospholipid transport **ATP-binding protein**
PP_0959	***mlaE***	Intermembrane phospholipid transport **permease**
PP_0960	***mlaD***	**Phospholipid-binding protein (this study, Q88P92)**
PP_0961	*ttg2D*	Toluene-tolerance protein
PP_0962	*ttg2E*	Toluene-tolerance protein
PP_0963	*ttg2F*	BolA-family protein

Two organism-specific conclusions follow:

1. **Genome-level validation of the transporter.** *mlaF* (ATPase) – *mlaE* (permease) – *mlaD* (binding subunit) are **adjacent and co-oriented**, confirming that *P. putida* MlaD assembles with MlaF and MlaE into the inner-membrane **MlaFED(B)** ABC transporter — validation by operon structure, not orthology alone. The neighboring LPS-biogenesis genes (*lptC*, *kdsC*, *kdsD*) reinforce a functional theme of **outer-membrane barrier construction and maintenance**.
2. **Direct link to *P. putida's hallmark solvent tolerance.** * The adjacent*** ttg2 ***genes are the*** ***toluene-tolerance** locus of *P. putida*. In *P. putida* GM73, the Ttg1/Ttg2 ABC transporters were identified as **required for organic-solvent (toluene) resistance**, with loss reducing toluene survival by more than six orders of magnitude; the authors concluded that "active efflux mechanism and efficient repair of damaged membranes were important in toluene resistance" [PMID 9658016]. Consistently, the **eggNOG orthologous group for Q88P92**

(COG1463) is annotated as an ABC-type transport system involved in **resistance to organic solvents**. Thus, in *P. putida* the Mla/Ttg2 machinery contributes to the organism's celebrated **solvent (toluene) tolerance**, most plausibly by maintaining OM lipid asymmetry and repairing solvent-induced membrane damage — an organism-specific physiological role that goes beyond the generic OM-asymmetry function.

8. Supported and Refuted Hypotheses

Supported 1. MlaD is the substrate-binding subunit (adapter, not enzyme) of the inner-membrane MlaFEDB ABC transporter. ✔ [PMID 34196044; 33199922] 2. MlaD binds glycerophospholipids and forms a homohexameric MCE ring with a hydrophobic central pore. ✔ [PMID 38347327; 39080293] 3. MlaD localizes to the inner membrane with its ring in the periplasm and directly interacts with the periplasmic shuttle MlaC. ✔ [PMID 37100290; 39080293] 4. The pathway drives ATP-dependent retrograde (OM→IM) phospholipid transport to maintain OM lipid asymmetry. ✔ [PMID 34873038] 5. In *P. putida*, *mlaD* is co-encoded with *mlaF*/*mlaE* in an operon linked to solvent (toluene) tolerance and OM/LPS homeostasis. ✔ [genomic context; P 9658016]

Refuted / not supported - MlaD as an *enzyme* catalyzing a chemical reaction — **not supported**; it is a binding/transport adapter. - MlaD as the ATPase or membrane permease of the complex — **not supported**; those roles belong to MlaF and MlaE, respectively.

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9. Limitations and Future Directions

- No *P. putida*-specific functional/structural study of PP_0960 was identified; the annotation rests on orthology to *E. coli* and related bacteria.
- The precise **directionality** (retrograde vs. bidirectional) and the exact **role of ATP hydrolysis** remain debated even in model organisms [PMID 39080293].
- **Future work:** solve a *P. putida* MlaFEDB/MlaD structure; characterize its native phospholipid cargo; and test *mlaD* deletion phenotypes (OM permeability, vesiculation, solvent/antibiotic tolerance) in KT2440 — relevant given *P. putida*'s notable solvent tolerance and envelope robustness.

Key References (PMID)

- 39080293 — Structure of the MlaC–MlaD complex (2024)
- 38347327 — Structural features of MlaD; hexameric MCE ring (2024)
- 34196044 — Conserved features of the MlaD domain (2021)
- 33199922 — Cryo-EM structures of MlaFEDB (2021)
- 34873038 — ATP drives retrograde transport (2021)
- 37100290 — MlaC–MlaD/MlaA interfaces; shuttle mechanism (2023)
- 35981415 — Structure/mechanism review of MlaFEDB (2022)
- 34753108 — Review: intermembrane lipid trafficking (2021)
- 37993432 — Ancestral Mla / transenvelope MlaD in *V. parvula* (2023)
- 36084896 — MlaC non-canonical SBP, polyspecific PL binding (2022)
- 31870848 — Cryo-EM of YebT (MlaD homologue) (2020)
- 29986943 — Mla pathway essential for intrinsic resistance in *Burkholderia cepacia* complex (2018)
- 9658016 — Toluene-sensitive (*ttg*) mutants of *P. putida*; Ttg2 ABC transporter required for solvent tolerance (1998)

Sequence source: UniProt Q88P92 (PP_0960; RefSeq WP_010952152.1); genome references Nelson et al. 2002 (P 12534463) and Belda et al. 2016 (P 26913973).